

Stage 2-2 - Clustering Analysis

Project: Customer Churn Analysis for Telecommunication Company

Team: WILDA Team 6

1. Objective

This analysis identifies distinct customer segments using K-Means clustering. Customer characteristics are used to form the clusters. The known Churn outcome is excluded from the clustering features and reviewed only after the model is trained to support interpretation.

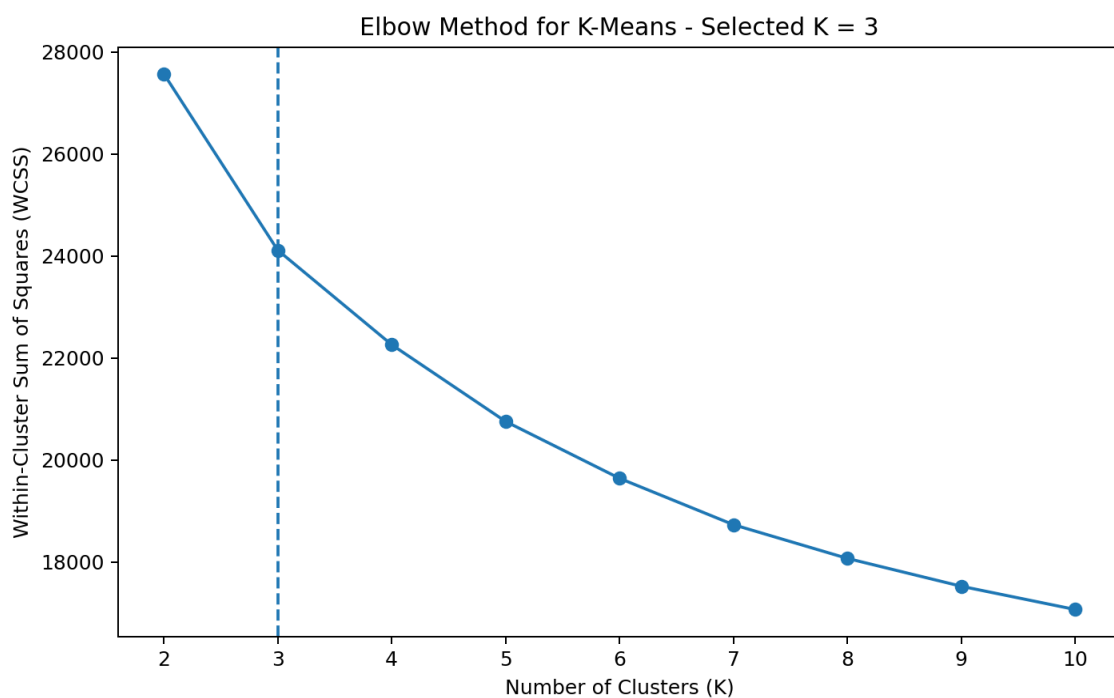
2. Feature Preparation

Nominal categorical variables were one-hot encoded. Tenure and MonthlyCharges were standardised using StandardScaler because K-Means relies on Euclidean distance. SeniorCitizen was retained as a binary 0/1 indicator. All 7,043 records were included.

Feature group	Treatment
gender, Dependents, PhoneService, MultipleLines, InternetService, Contract	One-hot encoded
tenure, MonthlyCharges	Standardised
SeniorCitizen	Binary 0/1 retained
Churn	Excluded from K-Means; used only for interpretation

3. Optimal Number of Clusters

K values from 2 to 10 were evaluated using the elbow method. The largest change in the WCSS slope occurs at K = 3. After K = 3, the reduction in WCSS becomes progressively smaller, indicating diminishing improvement from additional clusters.



A silhouette score was also calculated as a supporting diagnostic. It was highest at K = 2. This indicates that the dataset does not contain extremely separated natural groups. However, the assignment specifically requires elbow-method selection, and K = 3 represents the strongest elbow while providing three useful and interpretable customer segments.

4. Final K-Means Model

The final model was trained with KMeans(n_clusters=3, random_state=42, n_init=20). A fixed random state makes the result reproducible, while multiple initialisations reduce sensitivity to an unfavourable starting centroid.

5. Customer Segment Profiles

Cluster	Segment label	Customers	Avg tenure	Avg charge	Churn
0	Lower-Cost Lower-Churn Customers	2278	29.1 mo	\$27.96	13.0%
1	Newer Higher-Churn Customers	2534	12.7 mo	\$75.81	48.3%
2	Established Higher-Cost Lower-Churn Customers	2231	58.0 mo	\$89.78	15.6%

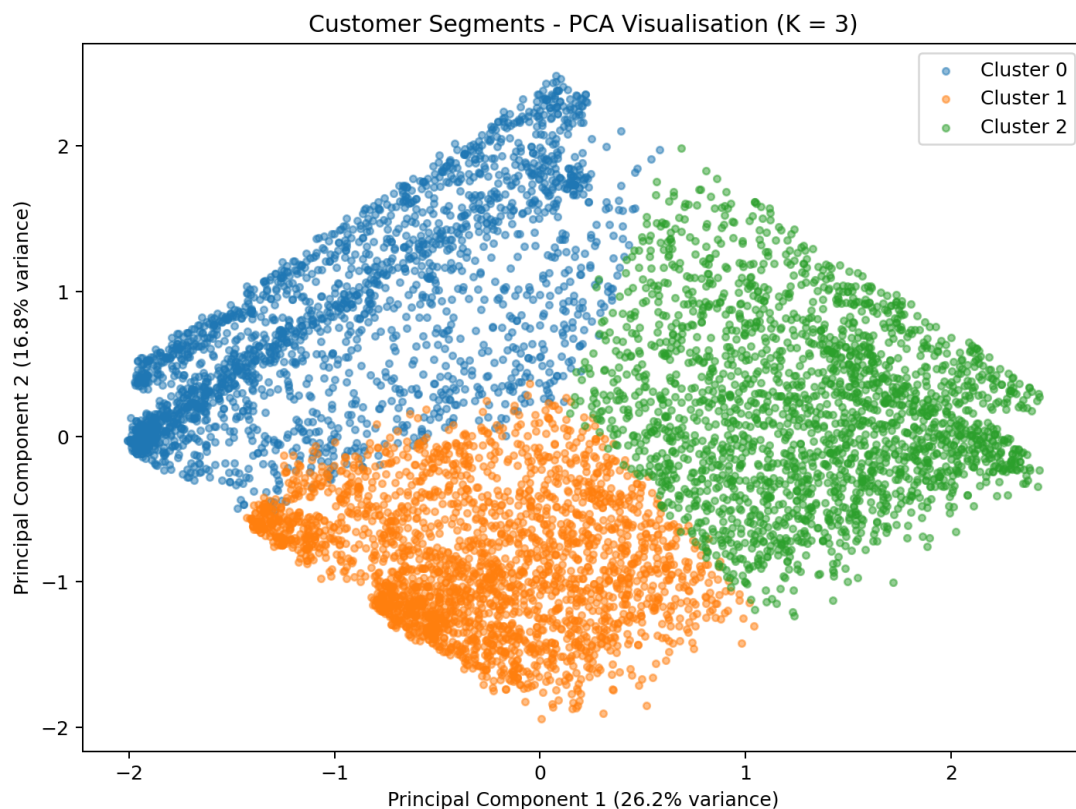
Cluster 0 - Lower-Cost Lower-Churn Customers: 2278 customers; average tenure 29.1 months; average monthly charge \$27.96; observed churn 13.0%. This comparatively stable group has the lowest average monthly charges and below-average observed churn. The business can focus on maintaining satisfaction and carefully selected value-added offers.

Cluster 1 - Newer Higher-Churn Customers: 2534 customers; average tenure 12.7 months; average monthly charge \$75.81; observed churn 48.3%. This group has the shortest average tenure and the highest observed churn. It is the clearest retention-priority segment, particularly for onboarding, early engagement, value reviews, and appropriate contract-conversion offers.

Cluster 2 - Established Higher-Cost Lower-Churn Customers: 2231 customers; average tenure 58.0 months; average monthly charge \$89.78; observed churn 15.6%. This group has the longest average tenure and highest average monthly charges while maintaining relatively low observed churn. Protecting loyalty and service quality is important because these are established higher-value customers.

6. Visualisation

PCA was used only to project the full clustering feature space into two dimensions for visual inspection. PCA was not used to fit the K-Means model.



7. Validation and Conclusion

All 7,043 customer records received exactly one cluster assignment. No missing or infinite values were introduced. Churn was excluded from the model input. The final saved model, cluster assignments, profiles, charts, code, and documentation are included in this Stage 2-2 package.